Scalable Fraud Detection: A Concurrent NLP System with Active Learning Integration

Unit: Object-Oriented Programming (OOP)
Focus: Milestones 5 & 6 (Concurrency & Innovation)
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Abstract: *This paper presents a high-performance NLP system designed for the detection of fraudulent job postings. By evolving the architecture from a synchronous model to a concurrent system utilizing Python's* `multiprocessing` *(Milestone 5), we achieve significant throughput gains. Furthermore, the system incorporates an Active Learning mechanism (Milestone 6) to flag uncertain predictions for manual review, demonstrating a research-level approach to iterative model improvement within an Object-Oriented framework.*

1. Introduction

In modern cybersecurity, the speed of processing is as critical as the accuracy of the model. This project addresses the challenge of scaling a Python-based NLP pipeline for large datasets. By applying advanced OOP principles, we ensure that the system remains modular while utilizing the full computational power of multi-core processors.

2. Object-Oriented Design & Custom Exceptions

The system adheres to core OOP principles to manage complexity. The `NLPSystem` class encapsulates the training and prediction logic, while the `TextPreprocessor` utilizes the Strategy Pattern to allow for flexible cleaning routines. Robustness is maintained through a custom exception hierarchy, starting with a base ***NLPError*** class, ensuring that data load failures or model state errors are handled gracefully in a concurrent environment.

3. Milestone 5: Concurrency and Performance Optimization

To overcome the limitations of the Global Interpreter Lock (GIL), the system was refactored to use `concurrent.futures.ProcessPoolExecutor`. This allows the CPU-intensive text cleaning and TF-IDF vectorization to be distributed across all available CPU cores. Performance benchmarks show a near-linear speedup in data processing time compared to the previous sequential implementation.

4. Milestone 6: Research Contribution - Active Learning

The research innovation for this milestone is the implementation of **Active Learning via Uncertainty Sampling**. Instead of accepting all predictions at face value, the system calculates the prediction probability $P(y|x)$. Samples where the model's confidence is low (e.g., $0.4 < P < 0.6$) are isolated as "Uncertain." This allows for a human-in-the-loop workflow where the model identifies its own weaknesses to guide future data labeling and training.

5. Conclusion

The final system demonstrates a complete transition from a basic script to a research-standard software tool. By integrating concurrency and intelligent automation, the project meets the high-performance requirements of Milestone 5 and the innovative research standards of Milestone 6, providing a robust solution for real-world fraud detection.

References

- Python Software Foundation. (2026). *Concurrency in Python: Multiprocessing and AsyncIO*.
- Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*.
- Settles, B. (2009). *Active Learning Literature Survey*. University of Wisconsin-Madison.